MAJOR PROJECT REPORT

ON

Stridex Marathon App: Enhancing Marathon Coordination Through Technology

Submitted By

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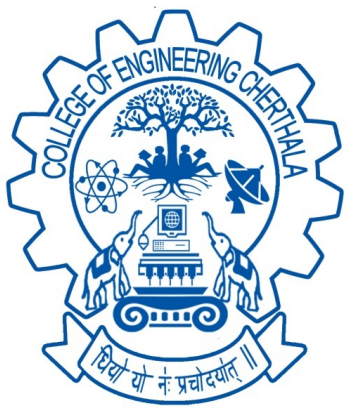
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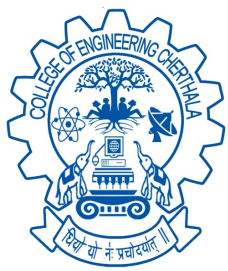
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In partial fulfillment of the requirements for the award of the degree
of
Bachelor of Technology
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C E R T I F I C A T E

This is to certify that, the project report titled ***STRIDEX MARATHON APP: ENHANCING MARATHON COORDINATION THROUGH TECHNOLOGY*** is a bonafide record of the **CSD416 Project Phase II** presented by **Abikrishna S Saiju**(Reg.No. CEC22CS011),**Alayna Margret Johny**(Reg.No. CEC22CS024) and **Aleena Thomas**(Reg.No. CEC22CS025),**Akhila P K** (Reg.No. LCEC22CS132), Eighth Semester B.Tech. Computer Science and Engineering student, under our guidance and supervision, in partial fulfillment of the requirements for the award of the degree, **B. Tech. Computer Science and Engineering** of **APJ Abdul Kalam Technological University**.

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ABSTRACT

Stridex — All in One App is an innovative mobile application developed using Flutter for the frontend and Firebase for the backend, designed to streamline and modernize marathon management through a single integrated platform. The app features GPS-based geofencing checkpoints and live runner tracking, allowing organizers to monitor participants in real time and analyze their performance through live statistics such as distance, pace, and timing. Users can easily create and register for marathons, manage bidding and sponsorships, and access an integrated online store to purchase event merchandise and essentials. The platform also supports volunteer registration, ensuring smooth coordination among event staff and participants. Furthermore, the built-in emergency "SOS" feature enables runners to instantly alert organizers with their live location during emergencies, ensuring timely assistance and enhanced safety. By combining technology, fitness, and event management, Run, Track, Engage provides a complete solution that connects participants, organizers, and sponsors, making marathon events more interactive, efficient, and secure.

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Chapter 1

INTRODUCTION

Marathon events have evolved from traditional physical gatherings into highly organized, technology-driven experiences that require real-time coordination, monitoring, and communication between organizers, participants, and volunteers. Managing such large-scale events involves multiple challenges, including participant registration, live tracking, safety management, sponsorship handling, and event promotion. Conventional methods often rely on manual tracking or fragmented systems, which can lead to inefficiency, data inaccuracy, and lack of engagement among participants.

To overcome these limitations, the project "Stridex — All in One App" introduces a unified mobile platform that digitizes and simplifies every aspect of marathon management [1]. Developed using Flutter for the frontend and Firebase for the backend, the application provides a seamless and cross-platform user experience. It integrates advanced GPS-based geofencing checkpoints and real-time runner tracking, allowing event organizers to monitor participants' positions accurately throughout the race[2],[3],[13]. The system also offers live performance analytics, such as pace, distance covered which help runners and organizers track progress effectively.

Beyond tracking, the application enables users to create and register for marathons, manage bidding and sponsorships, and operate an integrated online store for merchandise and race essentials. It also supports volunteer registration, ensuring efficient event staffing and coordination. Moreover, an emergency "SOS" feature is integrated to ensure runner safety by instantly notifying organizers with the runner's live location in case of emergencies.

In today's digital era, mobile technology plays a vital role in connecting individuals and enhancing their experiences in sports and fitness activities. By leveraging cloud-based infrastructure and real-time data analytics, Stridex bridges the gap between technology and sports. It ensures a smoother, safer, and more engaging marathon experience for everyone involved — from participants and volunteers to sponsors and organizers. Ultimately, the app aims to create a smarter, more interactive, and health-conscious community around marathon events, redefining how such large-scale athletic gatherings are managed and experienced.

Chapter 2

PROBLEM STATEMENT

2.1 Problem Statement

Organizing and managing marathon events involve numerous complex tasks such as participant registration, live tracking, volunteer coordination, sponsorship management, and safety monitoring. Traditionally, these processes are handled through multiple disconnected systems or manual methods, which often lead to inefficiencies, communication gaps, and inaccurate data handling[1]. Event organizers struggle to monitor runners in real time, ensure their safety, and provide timely updates to spectators and participants. Likewise, participants lack access to integrated tools that offer performance insights, progress tracking, and emergency assistance during the marathon[7].

Furthermore, the absence of a unified digital platform makes it difficult to manage sponsorships, event promotions, and post-event engagement [4],[8]. This lack of integration results in delayed decision-making, reduced participant satisfaction, and poor overall event coordination. With the growing scale of marathons and public fitness events, there is an urgent need for a smart, centralized, and technology-driven solution that connects organizers, runners, volunteers, and sponsors within a single ecosystem.

2.2 Objective

The main objective of the project "Stridex — All in One App" is to develop a unified and intelligent mobile application that simplifies marathon management and enhances the experience for organizers, participants, and volunteers. The system aims to integrate real-time tracking, event coordination, safety features, and engagement tools within a single platform.

1. To implement GPS-based geofencing and live runner tracking, enabling organizers to monitor participants' real-time locations, verify checkpoint completions, and ensure accurate event supervision.

2. To design a secure and efficient registration system, allowing users to create, manage, and participate in marathon events seamlessly within the application.

3. To integrate a sponsorship and bidding management feature, simplifying the process of connecting sponsors with event organizers and ensuring transparent financial handling.

4. To include an integrated online store, where participants can purchase marathon-related merchandise, accessories, and other essentials conveniently through the app.

5. To enhance overall event coordination and participant experience, by combining tracking, health insights, safety, and engagement tools into a single comprehensive platform.

Chapter 3

LITERATURE SURVEY

3.1 [1] Real-Time Location Tracking Using Browser-Based Geolocation API

Authors: Manav Jha, Abhinav Kale, Pratik Golder, Sahil Badole, and Vrushali Awale Published On: 2024

- **About:** This paper presents a browser-based system for real-time location tracking without requiring a dedicated mobile application.
- **Methodology:** The system uses the **HTML5 Geolocation API** with **Google Maps integration** to track user location dynamically using the **watchPosition()** method.
- **Advantages:** Platform-independent and easy to implement using standard web technologies. Provides live visualization of user location.
- **Disadvantages:** Limited GPS accuracy, requires continuous internet connectivity, and lacks advanced features like geofencing and offline support.

3.2 [2] GPS-Based Location Monitoring System with Geo-Fencing Capabilities

Authors: A. H. Abbas, Mohammed I. Habelalmateen, Syukran Jurdi, L. Audah, and N. A. M. Alduais Published On: 2019

- **About:** This paper proposes a GPS-based system for real-time tracking and geofencing alerts using IoT integration.
- **Methodology:** The system integrates **GPS modules, microcontrollers, and GSM/Wi-Fi communication** with cloud storage for monitoring and alerts.
- **Advantages:** Provides high accuracy and real-time alerts. Supports remote monitoring through cloud platforms.
- **Disadvantages:** Uses static geofencing zones, is focused on vehicle tracking, and lacks a user-friendly mobile interface.

3.3 [3] RFID Data Processing in a Real-Time Monitoring System for Marathon

Authors: Suchart Joolrat, Kanda Runapongsa Saikaew, Mongkon Tengrungroj, Patcharaporn Jiranuwattanawong, Channarong Janpanich, and Kittiphan Sornsakda Published On: 2016

- **About:** This paper presents an RFID-based system for monitoring marathon runners and calculating race results.
- **Methodology:** Uses **RFID tags, antennas, and readers** along with a web-based dashboard for tracking and result processing.
- **Advantages:** Provides high timing accuracy, reduces data processing time, and is scalable for large events.
- **Disadvantages:** Requires physical hardware setup, lacks continuous live tracking, and does not support mobile interaction.

3.4 [4] Geofencing 2.0: Taking Location-Based Notifications to the Next Level

Authors: Sandro Rodriguez Garzon and Bersant Deva Published On: 2014

- **About:** This paper introduces an advanced geofencing framework for intelligent location-based notifications.
- **Methodology:** Uses **state-transition geofencing** with temporal and duration-based rules for triggering notifications.
- **Advantages:** Enables context-aware and proactive notifications with flexible geofencing logic.
- **Disadvantages:** Complex implementation, requires technical expertise, and is not specifically designed for sports tracking.

3.5 [5] Real-Time Surveying and Monitoring of Athletes Using Mobile Phones and GPS

Authors: B. Eskofier, E. Hartmann, P. Kühner, J. Griffin, H. Schlarb, M. Schmitt, and J. Hornegger Published On: 2008

- **About:** This paper describes a mobile-based system for monitoring athlete performance and location using GPS-enabled phones.
- **Methodology:** Developed using **Java ME**, the system collects real-time data such as speed and movement patterns.
- **Advantages:** Lightweight design and validated through real-world testing. Provides useful performance tracking.

- **Disadvantages:** Based on outdated technology, lacks geofencing features, and has limited analytical capabilities.

Table 3.1: Literature Survey Summary

Ref	**Focus**	**Key Strength**	**Gap Identified**
[1]	Browser-based tracking	Platform independent, easy implementation	Low accuracy, no geofencing, internet dependent
[2]	GPS + Geo-fencing system	High accuracy, real-time alerts	Static zones, no mobile UI, vehicle-focused
[3]	Athlete monitoring using mobile GPS	Lightweight, tested with athletes	Outdated technology, no geofencing, limited analytics
[4]	Advanced geofencing system	Smart, context-aware notifications	Complex implementation, not suitable for sports tracking
[5]	RFID marathon tracking	Accurate timing, fast processing	No live tracking, requires hardware setup, no mobile support

Chapter 4

PROPOSED SYSTEM

4.1 Solution

The proposed system, Stridex, is a mobile application designed to simplify marathon management and enhance participant experience. Built with Flutter for cross-platform access and Firebase for real-time data and cloud storage, Stridex integrates multiple functionalities in a single platform.

The app features GPS-based geofencing and live runner tracking, allowing organizers to monitor participants in real time, while runners can view performance metrics like distance, pace, and best time. Users can create and register for marathons, manage sponsorships, and access an online store for merchandise[5],[10],[13].

For safety, Stridex includes an emergency "SOS" button to alert organizers with the live location of the runner.[7]. Overall, Stridex delivers a centralized, interactive, and secure platform that improves event coordination, participant engagement, and safety for marathon events.

1. User Authentication: Secure and fast login for organizers, participants, and volunteers.

2. GPS-Based Geofencing: Real-time tracking of runners at checkpoints throughout the marathon route.

3. Live Performance Analytics: Provides runners and organizers with distance, pace, and estimated finish time.

4. Event Registration and Management: Users can create, register, and manage marathon events seamlessly.

5. Sponsorship and Bidding Management: Facilitates sponsor participation and financial transparency for organizers.

6. Integrated Online Store: Purchase marathon merchandise, kits, and other essentials directly through the app.

7. Emergency SOS Feature: Allows runners to instantly alert organizers with their live location during emergencies.

8. Scalable System: Supports growing numbers of participants, events, and real-time data without performance issues.

Chapter 5

Feasibility Study

The objective of this study is to evaluate whether the proposed system can be successfully developed and implemented. For this purpose, the system is analyzed under three key feasibility considerations:

- Technical Feasibility
- Operational Feasibility
- Economic Feasibility

The feasibility of the Stridex system is assessed from multiple perspectives to ensure its practicality and effectiveness.

5.0.1 Technical Feasibility

The proposed system, Stridex, is technically feasible as it leverages Flutter for cross-platform mobile development and Firebase for real-time backend services, authentication, and cloud storage. Flutter ensures compatibility with Android devices, providing a responsive and consistent user interface. GPS tracking, geofencing, and real-time performance analytics can be efficiently implemented using existing APIs and Firebase services. The integration of biometric login, SOS alerts, and online payment modules is supported by standard mobile frameworks and cloud-based solutions, minimizing technical risks.

5.0.2 Operational Feasibility

Stridex is operationally feasible as it addresses the real needs of marathon organizers, participants, volunteers, and sponsors. Event organizers can efficiently manage registration, checkpoints, sponsorships, and communication, while runners receive real-time tracking, health insights, and safety alerts. Volunteers and admins benefit from centralized monitoring and coordination. The app's intuitive design ensures ease of use for all stakeholders, promoting adoption without extensive training or technical support.

5.0.3 Economic Feasibity

The development and deployment of Stridex are cost-effective due to the use of open-source frameworks (Flutter) and cloud services (Firebase) that reduce infrastructure and maintenance costs. The app eliminates the need for multiple fragmented systems for registration, tracking, and event management, lowering operational expenses for organizers. Additionally, the integrated online store and sponsorship management modules provide potential revenue streams, making the system economically sustainable.

Chapter 6

SOFTWARE REQUIREMENT SPECIFICATION

6.1 Overall Description

The Stridex application is a comprehensive marathon management system designed to modernize and unify the race experience for organizers and participants. The platform's purpose is to leverage mobile technology, using Flutter for the frontend and Firebase for the backend, to provide a centralized solution for event logistics and runner engagement. It aims to enhance operational efficiency for organizers through integrated tools for registration, sponsorship, and live monitoring, while simultaneously empowering runners with real-time GPS tracking, performance statistics, and a critical SOS safety feature. Through its Self-Marathon mode and personal progress analytics, the system also serves as a continuous training companion, promoting long-term user engagement beyond official events.

6.1.1 Project Perspective

Stridex is a self-contained marathon management and running companion platform built on a client–server architecture that integrates multiple components to ensure efficient event execution and a rich user experience. The web-based dashboard for event organizers and admins manages event creation, participant registration, sponsorships, and live monitoring of runners via a map-based interface. The mobile application for runners provides event discovery, registration, live GPS tracking during races, real-time performance statistics, and a critical emergency "SOS" but-

ton. A centralized Firebase backend stores user profiles, event data, transaction histories, and live location data, while the system utilizes the built-in GPS hardware in participants' smartphones for all tracking and location-based features. The system enhances the marathon experience by combining comprehensive event management with personalized runner engagement and safety, streamlining logistics for organizers while creating a more interactive and secure journey for participants.

6.1.2 Project Function

This project has many functions. They are listed below:

item Event creation, management, and participant registration.

- Live GPS tracking of all runners with geofenced checkpoints.
- Real-time performance statistics, including pace, distance, and calorie-burn estimation.
- Integrated Emergency "SOS" button to notify organizers with a live location.
- Management of sponsorships, bidding, and an integrated web store.
- Personal run logging (Self-Marathon mode) with progress tracking via interactive charts.

6.2 User Classes and Characteristics

- Runners / Participants: require a seamless mobile interface for event discovery, registration, live race tracking, and monitoring personal progress.
- Event Organizers / Admins: responsible for creating and managing events, overseeing participant registrations, monitoring runner safety in real-time, and analyzing event data.

6.2.1 Operating Environment

- Mobile Platforms: The runner's application is natively supported on both Android devices.

- Device: The mobile app is used on smartphones, while the management dashboard for organizers is accessed on desktops and laptops.
- Web Browser (for Admin Dashboard): The administrative interface is accessible via modern web browsers, including Chrome, Firefox, Safari, and Edge.
- Backend Environment: The entire system operates on Google's Firebase cloud platform, ensuring high availability and real-time data synchronization.

6.2.2 Design and Implementation Constraints

- Frontend Development: The primary mobile application for runners is developed using Flutter to ensure a consistent experience on both Android. The organizer's portal is a dedicated web-based dashboard.
- Backend Development: The backend is built on Google's Firebase platform, which handles user authentication, real-time data synchronization, and server-side logic using Cloud Functions.
- Database: The system uses Cloud Firestore,cloud-hosted database from Firebase, for scalable storage of all user profiles, event data, and live tracking information.
- Hardware Dependency: Core functionalities like live tracking, SOS alerts, and geofencing are critically dependent on the built-in GPS hardware of the user's smartphone.

6.2.3 Assumptions and Dependencies

- Users must have an active account and grant the application necessary permissions (e.g., location services, notifications).
- A stable internet connection is required for live tracking, data synchronization, and SOS alerts.

- The accuracy of performance analytics (pace, distance,best time) is dependent on the quality of the smartphone's GPS signal.

6.3 External Interface Requirements

6.3.1 User Interfaces

The system provides distinct user interfaces for the mobile app (for runners) and a web-based dashboard (for organizers). The mobile app is developed using Flutter to ensure a consistent cross-platform experience. Both interfaces are designed to be intuitive and tailored to the specific needs of their user classes.

6.3.2 Hardware Interfaces

The system's primary hardware interface is the smartphone's built-in GPS module, which is essential for all live tracking, geofencing, and safety features. The mobile app runs on standard Android and iOS devices, while the web dashboard is accessible from any desktop or laptop computer.

6.3.3 Software Interfaces

The system is built on the Google Firebase platform, using Cloud Firestore as its database. Communication between the mobile app, web dashboard, and the backend is handled in real-time via the Firebase SDKs. Server-side logic and triggers are managed by Firebase Cloud Functions.

6.4 Non-functional Requirements

- Performance: Runner location data must be updated and displayed on the organizers' live map in near real-time, with a latency of less than 5 seconds.
- Security: All user data, particularly live and historical location information, must be transmitted over secure channels and stored with encryption.

- Scalability: The system must be able to handle thousands of concurrent runners transmitting location data simultaneously during a large-scale marathon.
- Reliability: The core backend services must maintain at least 99.9
- Usability: The mobile app must feature a high-contrast, simple interface that is easily operable by a user in motion, with critical functions like the SOS button being immediately accessible.

Chapter 7

SYSTEM DESIGNS

7.1 Module

Modules used in this project are as below:

1. Event Management Module(Participant, Admin)

   When a participant opens the Stridex Marathon App and selects the "View Event" option, the system retrieves and displays all relevant event details, including the name, date, time, location, registration fee, and category. On the administrative side, the backend web panel allows the admin to create, edit, or delete marathon events as needed. Participants can then register for any available event, and upon successful registration, the system generates a unique Registration ID while securely storing the user's details. Finally, a confirmation of the registration is sent to the participant either through email or an in-app notification, ensuring they are informed and ready for the event.

2. Participant Registration Module(Participant, Admin)

   When a participant chooses the "Register for Event" option, the app prompts them to provide essential details such as Name, Age, Gender, Contact information, Category, and Payment. The system then validates both the entered data and the payment details to ensure accuracy and completeness. If the information is valid, the system securely stores the registration details and updates the participant's status to "Registered." Before the event day, the admin

reviews the registrations and verifies and approves participants, ensuring that only confirmed individuals are officially enrolled in the marathon.

3. Volunteer Registration Module(Participant, Admin)

   Participants who wish to volunteer can select the "Volunteer Registration" option within the app, where they are prompted to provide their role preferences and availability. Once submitted, the system records their details in the database for organizational purposes. The admin then has the ability to assign volunteers to specific responsibilities such as checkpoints, water stations, or logistics teams, ensuring smooth event management. Finally, a confirmation is sent to the volunteers, either via email or notification, to acknowledge their successful registration and assigned role.

4. Self Mode and Fitness Tracking Module(Participant)

   When a participant selects the "Self Mode" option, the system activates the device's GPS to track both distance and time during their run. Using this data, the app calculates the participant's pace. All performance data is stored locally on the device and then synced with the participant's profile for future reference. Participants can later access a detailed performance summary through the dashboard, allowing them to monitor progress and evaluate their training outcomes effectively.

5. Webstore Management Module (Participant, Admin)

   Participants can access the "Webstore" within the app to purchase marathon merchandise such as t-shirts, accessories, and energy bars. The system displays all available items along with their prices and current stock status, allowing participants to browse and select products of interest. Once they add the desired items to their cart and proceed to checkout, the system records the purchase details and updates the stock levels in the database accordingly. On the administrative side, the backend panel enables the admin to manage the store by adding new products, updating existing ones, or deleting items as needed, ensuring smooth operation of the merchandise store.

6. Live GPS Tracking Module (Participant, Admin)

   During the marathon, the participant's GPS is automatically activated through the app, enabling continuous real-time tracking of their location via integrated map APIs. This live data is transmitted to the admin dashboard, where organizers can monitor each participant's current position, route coverage, and running speed. By providing this visibility, the module ensures both safety and transparency throughout the event, allowing admins to respond quickly to any issues while maintaining accurate oversight of the marathon's progress.

7. Emergency SOS Module (Participant, Admin)

   If a participant encounters an emergency during the race, they can press the "SOS Button" in the app, which immediately transmits their current GPS coordinates along with their unique ID to the admin dashboard and the emergency response team. The admin then monitors the alert in real time and dispatches medical or on-ground assistance to the participant's location. Once the situation is resolved, the admin updates the system by marking the case as "Handled," ensuring proper tracking and closure of emergency incidents during the marathon.

8. Bidding and Sponsorship Module (Participant(Sponsor), Admin)

   Admins can publish available sponsorship opportunities and stall spaces through the web portal, making them visible to both participants and external parties. Interested sponsors can then place bids or directly contact the admin via the app to express their interest. Once a sponsorship request is received, the admin reviews and approves it, after which the system updates sponsor details such as name, contribution amount, and brand logo on event materials. To ensure proper documentation, billing information or receipts are automatically generated for each approved sponsor, streamlining the sponsorship process and maintaining transparency.

7.2 System Architecture

The Stridex Marathon Management System uses a modular, cloud-based architecture to ensure efficient and scalable event management. It integrates a mobile application, web dashboard, GPS tracking, and cloud services into a single platform.

A centralized Firebase cloud database stores all data, including participant details, event information, and real-time tracking data. The mobile application (Flutter) is used by runners and volunteers for registration, tracking, and receiving updates, while the web dashboard allows organizers to manage events and monitor activities.

The system uses GPS technology for real-time tracking of runners and geofencing to ensure they follow the correct route. An SOS feature is included for emergency alerts with live location sharing.

Overall, the architecture provides a reliable, scalable, and real-time solution that improves coordination, safety, and efficiency in marathon events.

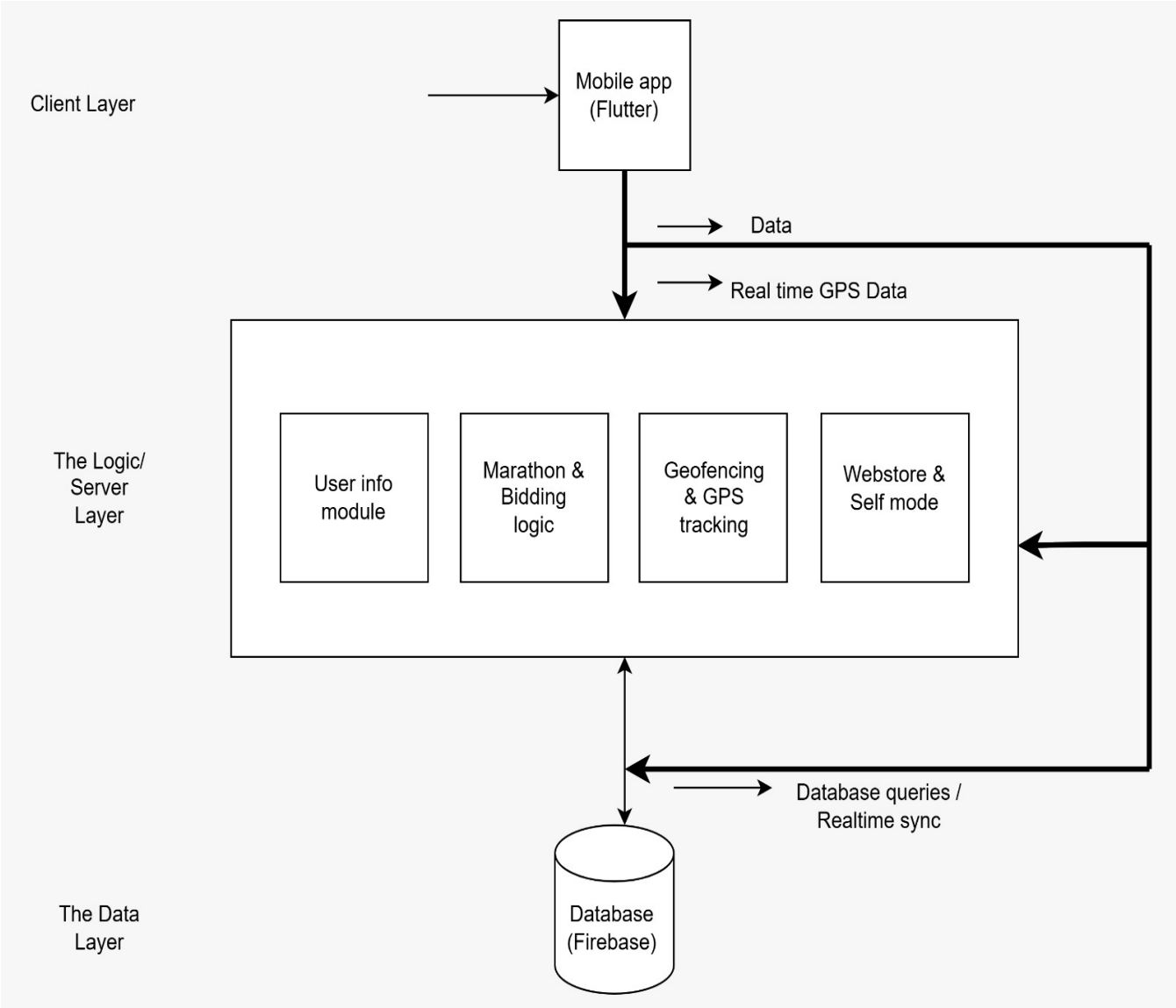


Fig. 7.1: System Architecture

7.3 Use-case Diagram

A use case diagram represents the functional requirements of a system by showing the interactions between users (actors) and the system. It provides a high-level view of system functionalities and user roles.In the proposed System, three primary actors are identified:

- Participants
- Admin
- sponsors

1. The participant is the end-user who registers for and takes part in marathon events through the mobile application.They interact with the system to view events, track performance, and receive real-time updates during the marathon.

2. The admin is responsible for managing and controlling all marathon operations, including events, registrations, and sponsors.They ensure smooth functioning of the system by monitoring activities, handling emergencies, and maintaining data accuracy.

3. Sponsors interact with the system to promote their brand during marathon events. They can participate in bidding for sponsorship slots, manage advertisements, and track their promotional activities. The system provides a platform for sponsors to engage with events and gain visibility among participants and spectators.



Fig. 7.2: Usecase diagram

7.4 ER Diagram

The Marathon Management System is modeled through an ER diagram that represents its core entities and their relationships within the Firebase database. Organizers are responsible for creating and managing marathon events, setting details such as date, route, and registration fees, and assigning bib numbers to runners. Runners can register for available marathons, participate in live races, and share GPS tracking data in real time during the event. Spectators are connected to the system by viewing live tracking information, race progress, and final results, which enhances engagement and transparency. The system also records and manages all live GPS data, race results, and performance statistics such as pace, distance, and best time. Additionally, the Webstore entity allows organizers to manage merchandise while enabling runners to purchase event-related items. The SOS entity is linked to both runners and organizers, ensuring immediate alerts with real-time location data during emergencies. After the event, organizers can generate performance reports, while runners can access their race history through the interconnected database relations.

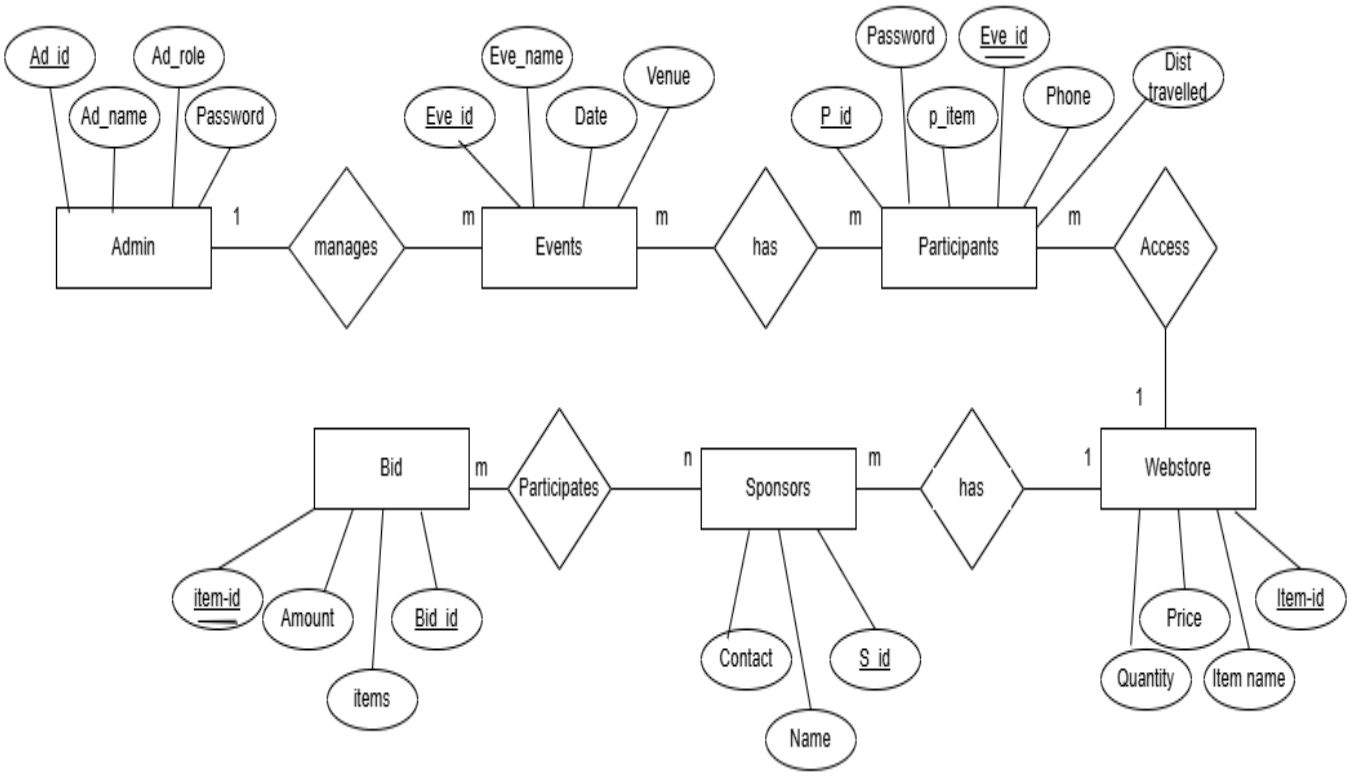


Fig. 7.3: ER Diagram

7.5 Sequence Diagram

The Sequence Diagram represents the interaction between the main actors—Runner, Organizer, Sponsor, and System—in a time-ordered manner. It illustrates how the runner registers and starts the race, while the organizer creates and manages events. During the marathon, the system continuously processes GPS data to provide real-time tracking and updates. The sponsor interacts with the system to manage advertisements. Overall, the diagram shows the sequential flow of communication within the system.

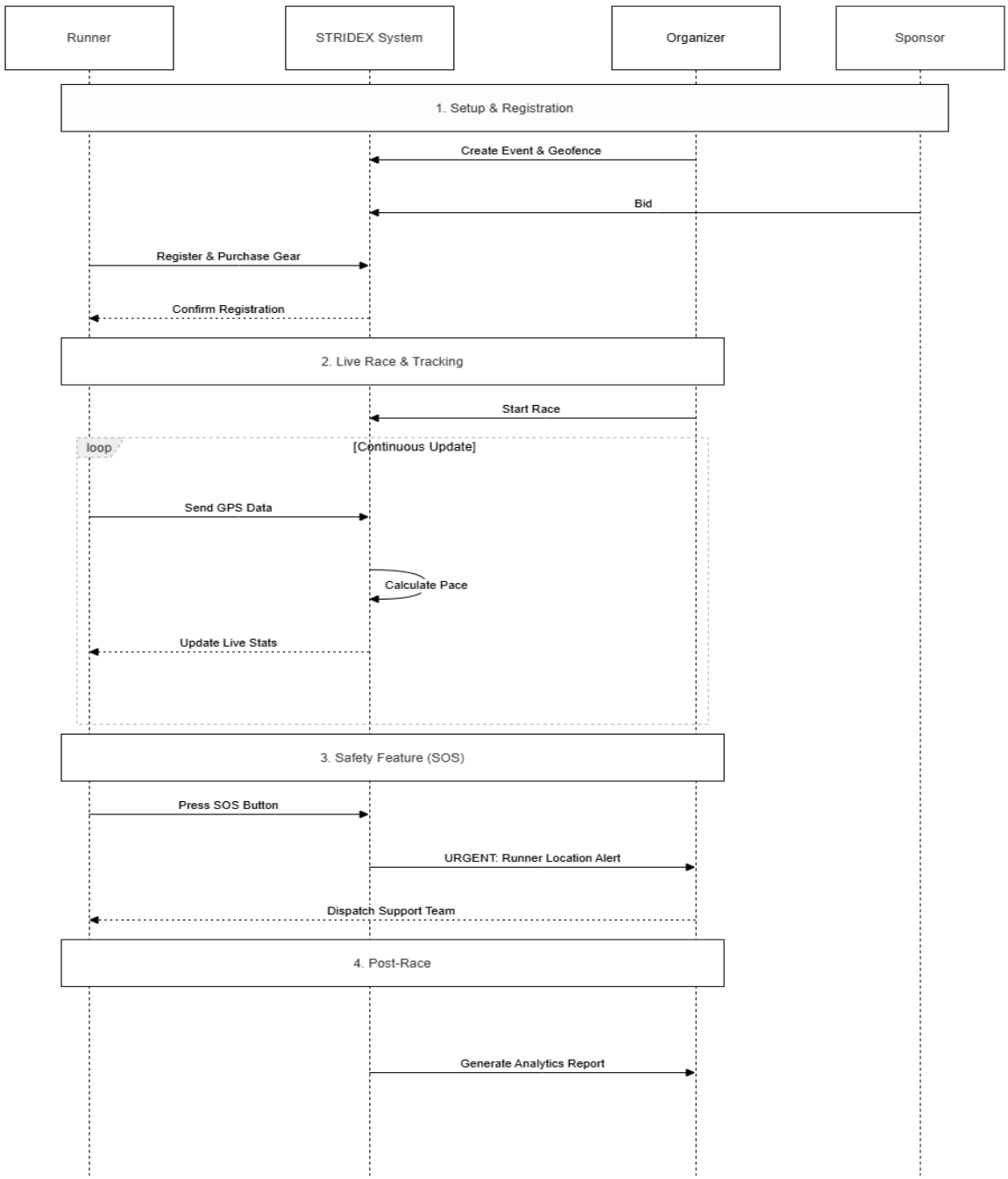


Fig. 7.4: Sequence Diagram

7.6 Data Flow Diagram

The data flow diagram (DFD) is used for classifying system requirements to major transformation that will become programs in system design. This is starting point of the design phase that functionally decomposes the required specifications down to the lower level of details

Bubbles: Represent the data transformations.

Lines: Represent the logic flow of data.

Data can trigger events and can be processed to useful information. Systems analysis recognizes the central goal of data in organizations.

7.6.1 Level 0

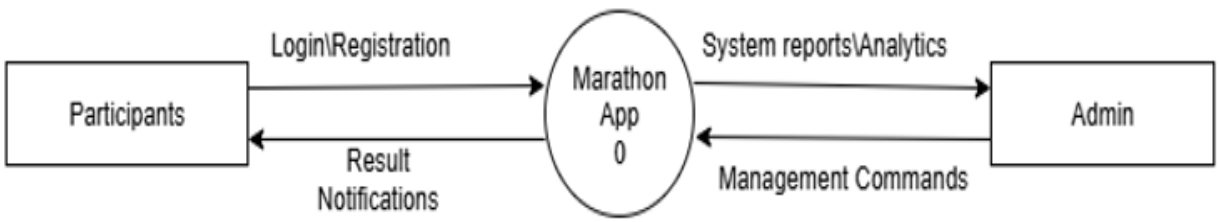


Fig. 7.6: Level 0 DFD

7.6.2 Level 1

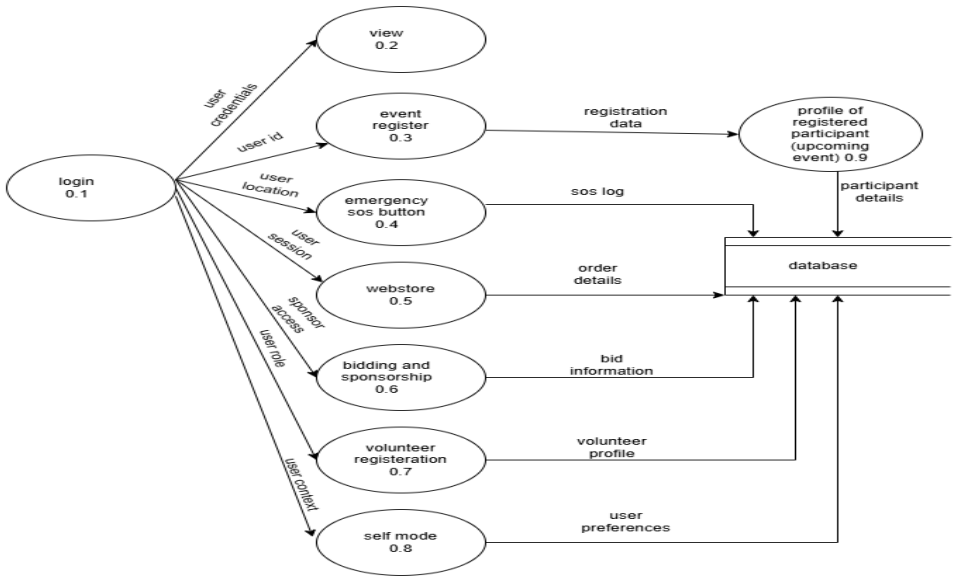


Fig. 7.7: level 1 DFD

7.6.3 Level 2

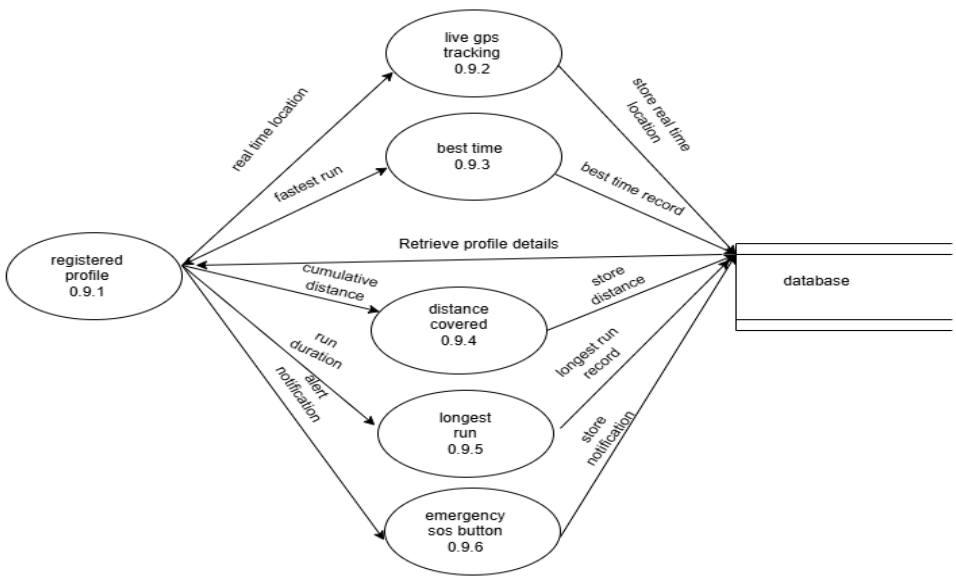


Fig. 7.8: level 2 DFD

7.6.4 Level 3

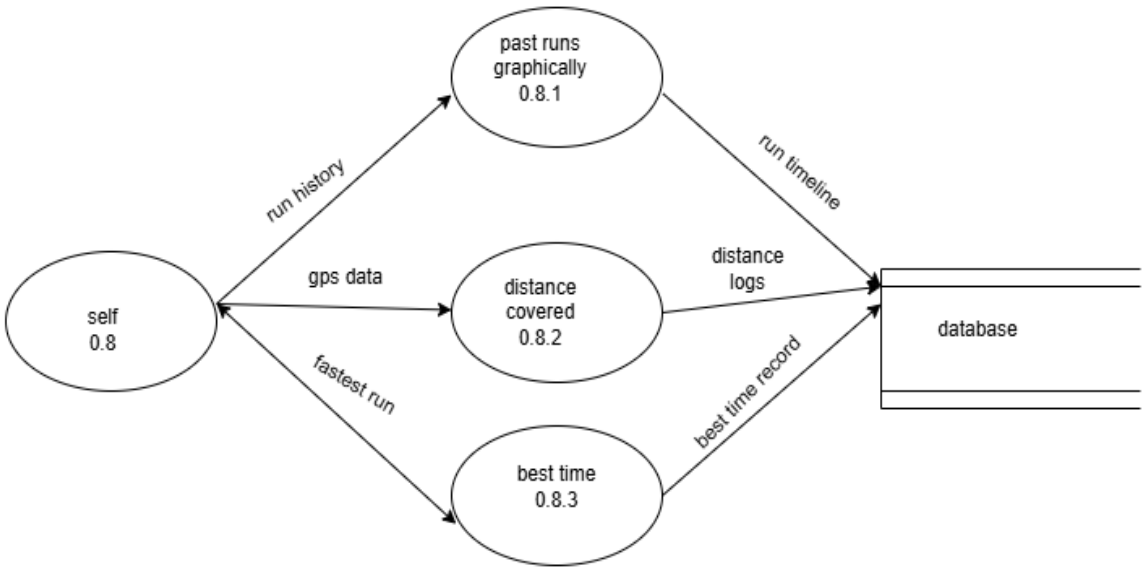


Fig. 7.9: level 3 DFD

Chapter 8

SOFTWARE AND HARDWARE REQUIREMENT

8.1 Software Requirements

- Frontend : DART
- Backend: Google Firebase (including Firestore, Authentication, and Cloud Functions)
- Mobile Application: Flutter
- Mapping Location SDK/API: Google Maps API and native iOS/Android location services.

8.2 Hardware Requirements

- The primary hardware is the user's smartphone, which must have a functional GPS module.
- Event organizers access the system using PCs or laptops, while runners use their smartphones.

8.2.1 Laptop Requirements (for Development/Organizers)

- Processor: 12th GEN INTEL, i5 or equivalent
- Main memory: 16GB
- Minimum CPU: 2 cores (2.30GHz)

- Minimum RAM: 4GB and above
- Minimum Storage: 50GB

Chapter 9

IMPLEMENTATION

9.1 System Overview

The Stridex Marathon Management System is designed to simplify and optimize the management of marathon events through a unified digital platform. The system integrates key functionalities such as event creation, participant registration, real-time GPS tracking, volunteer coordination, sponsorship management, and performance analytics. It provides a centralized environment where organizers, runners, volunteers, and sponsors can interact efficiently. By combining mobile and cloud technologies, the system enhances coordination, improves safety, and delivers a user-friendly experience for all stakeholders[5].

9.2 Main Components

9.2.1 User Management

The User Management component handles authentication and user roles within the system. Users such as runners, volunteers, and administrators can register using their personal details. Secure login is provided through authentication mechanisms, ensuring data privacy. The system supports role-based access control, allowing each user type to access only relevant features. Users can also manage their profiles, update information, and view their activity history[1],[2],[5].

9.2.2 Event Management

The Event Management component allows administrators to create and manage marathon events. Organizers can define event details such as date, location, route, categories, and registration fees. These details are stored in a centralized cloud database and made accessible to users through the mobile application. The system ensures proper coordination of event-related activities and provides real-time monitoring capabilities for organizers.

9.2.3 Registration System

The Registration System enables runners and volunteers to register for events and efficiently. Users can sign up by providing personal details, and the system generates unique identifiers for each participant. For paid events, secure payment processing is supported. The system maintains registration records and provides confirmation notifications to users. Volunteers can also register and select roles for event participation.

9.2.4 Tracking and Safety Module

This module is responsible for real-time GPS tracking and safety features. During the marathon, the system continuously tracks the location of runners using GPS and displays it on the organizer's dashboard. Geofencing is used to ensure participants follow the designated route. The system also includes an SOS feature that allows runners to send emergency alerts along with their live location, enabling quick response from organizers.

9.2.5 Analytics and Reporting

The Analytics and Reporting component processes data collected during events to generate insights. It provides real-time statistics such as distance covered, pace, and performance metrics. The system also generates leaderboards and post-event reports. These reports help organizers evaluate event performance and improve future events.

9.3 How It Works

9.3.1 For Runners

Runners interact with the system through a mobile application. They can register for events, view event details, and track their performance in real time. During the marathon, GPS tracking provides live updates on their progress. Runners can also monitor metrics such as distance, pace, and calories burned. In case of emergencies, they can use the SOS feature to alert organizers.

9.3.2 For Organizers

Organizers use a web-based dashboard to manage events and monitor activities. They can create events, manage registrations, coordinate volunteers, and track participants in real time. The dashboard provides a complete overview of the event, including live tracking data and emergency alerts, enabling efficient decision-making.

9.3.3 For Volunteers

Volunteers register through the system and assist in managing event operations. They help with tasks such as monitoring checkpoints, supporting runners, and coordinating with organizers. The system allows volunteers to view assigned duties and communicate effectively during the event.

9.3.4 For Sponsors

Sponsors interact with the system through sponsorship modules. They can bid for promotional opportunities and manage advertisements. The platform provides visibility and engagement opportunities for sponsors, enhancing event profitability.

9.4 Technical Setup

The technical setup of the system is based on modern mobile and cloud technologies. The application is developed using Flutter for cross-platform compatibility, while Firebase is used as the backend for authentication, database, and cloud storage. Google Maps API is integrated for GPS tracking and route visualization. The system ensures real-time data synchronization and efficient handling of large-scale event data.

9.4.1 Database Tables

The system uses multiple database collections to manage data efficiently. The Users collection stores information about runners, volunteers, and administrators. The Events collection maintains event details such as schedules and routes. The Registrations collection records participant data. The Tracking data stores real-time GPS information. Additional collections are used for sponsorships, payments, and performance analytics. These components ensure structured data storage and quick retrieval.

9.4.2 Key Features Working Together

The system operates through coordinated interaction between its components. Organizers create events, which are stored in the cloud database. Users register for events through the mobile app, and their data is synchronized in real time. During the event, GPS tracking continuously updates runner locations, and geofencing ensures route compliance. After the event, analytics and reports are generated based on collected data. This integration ensures smooth and efficient marathon management.

9.5 Security Features

Security is an important aspect of the system. User authentication ensures that only authorized users can access the platform. Role-based access control restricts system functionalities

based on user roles. Data is securely stored in the cloud, and sensitive information is protected from unauthorized access. The system also ensures secure handling of user data and communication, maintaining privacy and reliability throughout the platform.

Chapter 10

RESULT AND ANALYSIS

10.1 Screenshots

10.1.1 Home

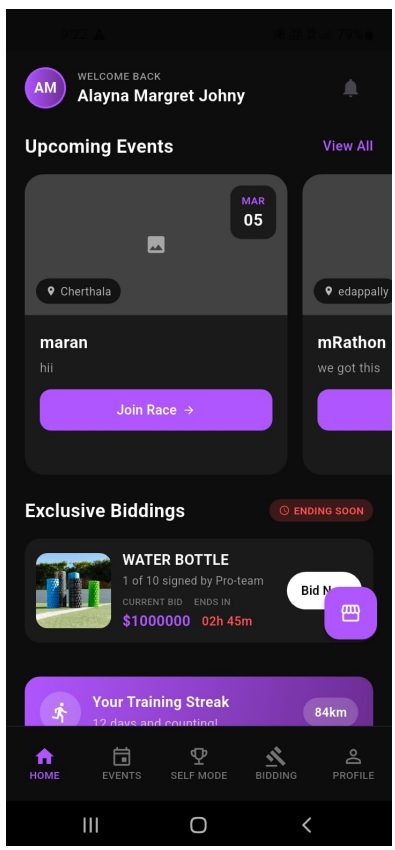


Fig. 10.1: Home
It's a fitness app dashboard showing events, bidding, and training streak.

10.1.2 Login

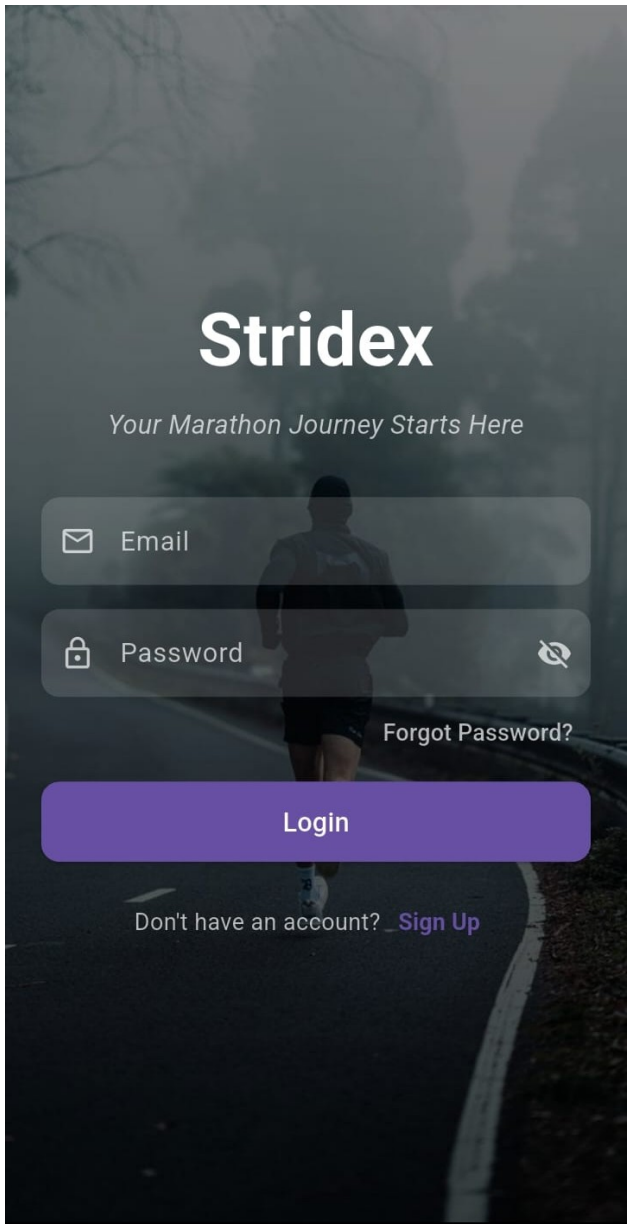


Fig. 10.2: Login
It's a Stridex app login screen with email, password, and sign-up options.

10.1.3 Signup

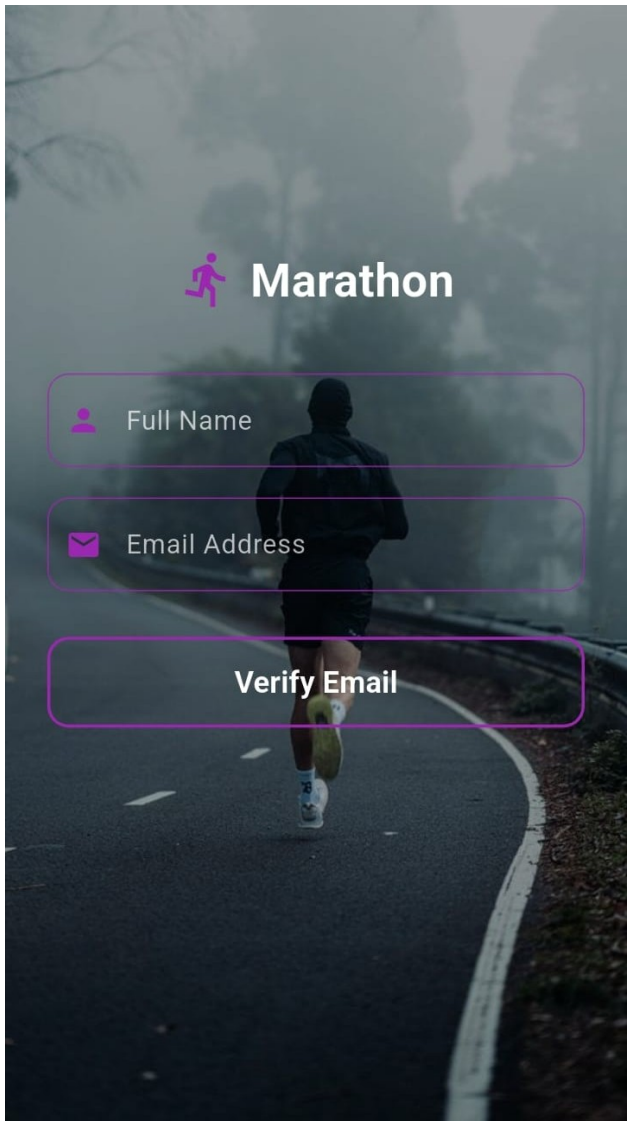


Fig. 10.3: Signup
It's a marathon registration screen with fields for name, email, and a verify button.

10.1.4 Events page

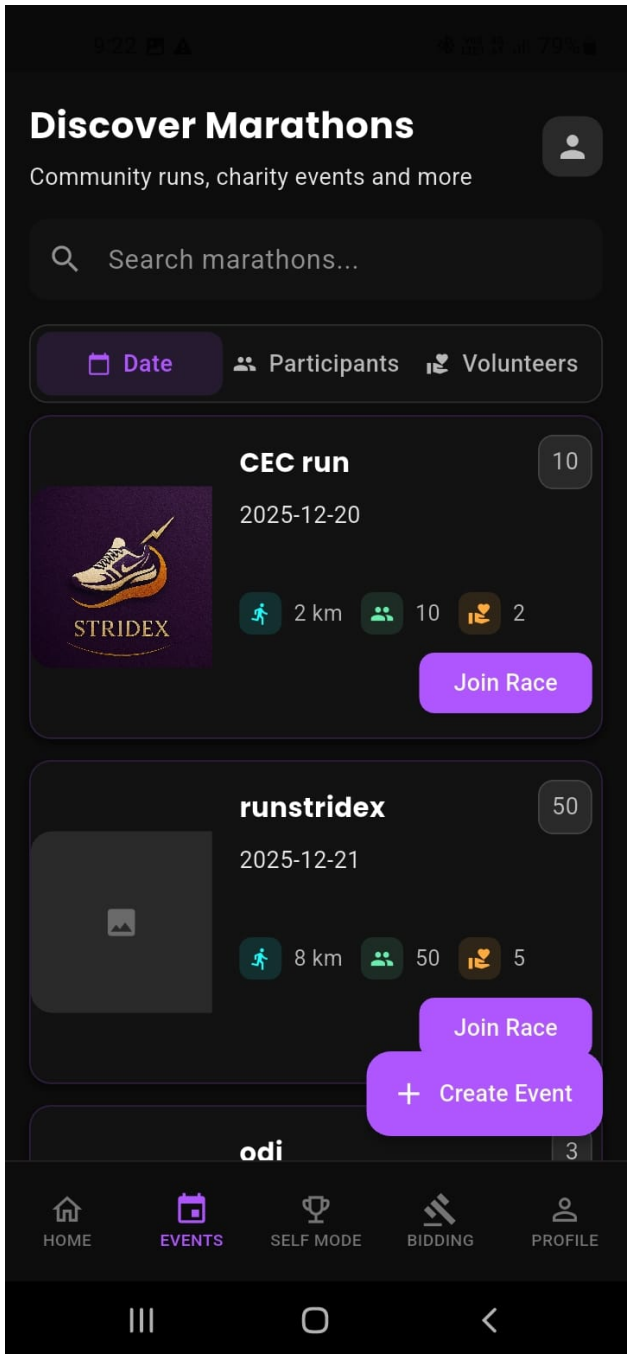


Fig. 10.4: events page

It's a marathon discovery screen showing upcoming races with details like date, distance, participants, volunteers, and options to join or create events.

10.1.5 webstore

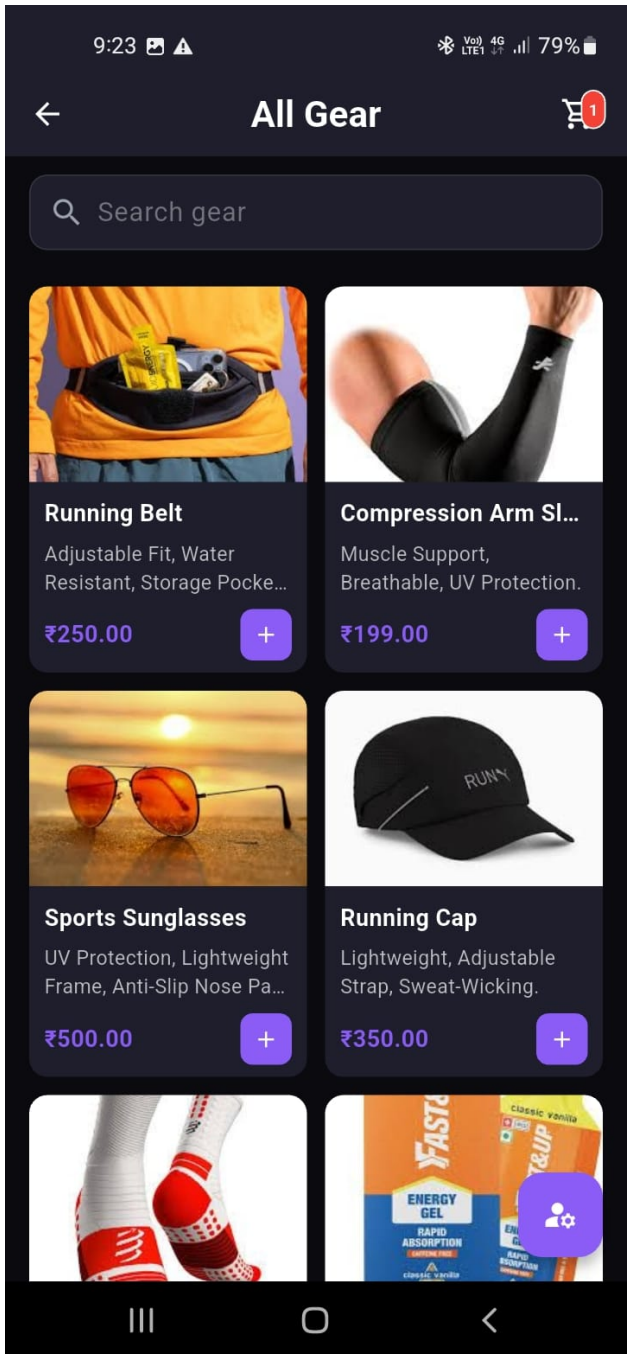


Fig. 10.5: webstore

It's a gear shop screen showing running accessories with prices and add-to-cart options.

10.1.6 self page



Fig. 10.6: self page

It's a self-tracking screen showing GPS status, run metrics, a weekly activity graph, and an AI coach chat option.

10.1.7 Bidding

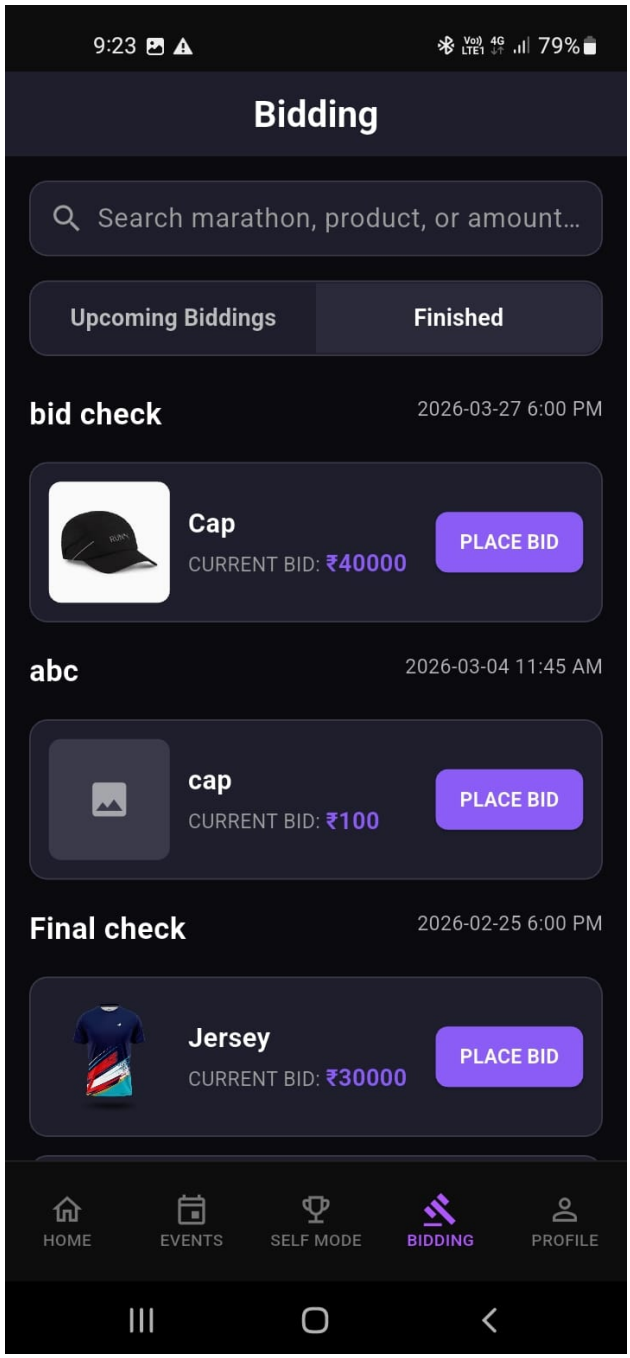


Fig. 10.7: Bidding

It's a bidding screen showing finished auctions with items like caps and jerseys, their current bids, dates, and options to place a bid.

10.1.8 profile page

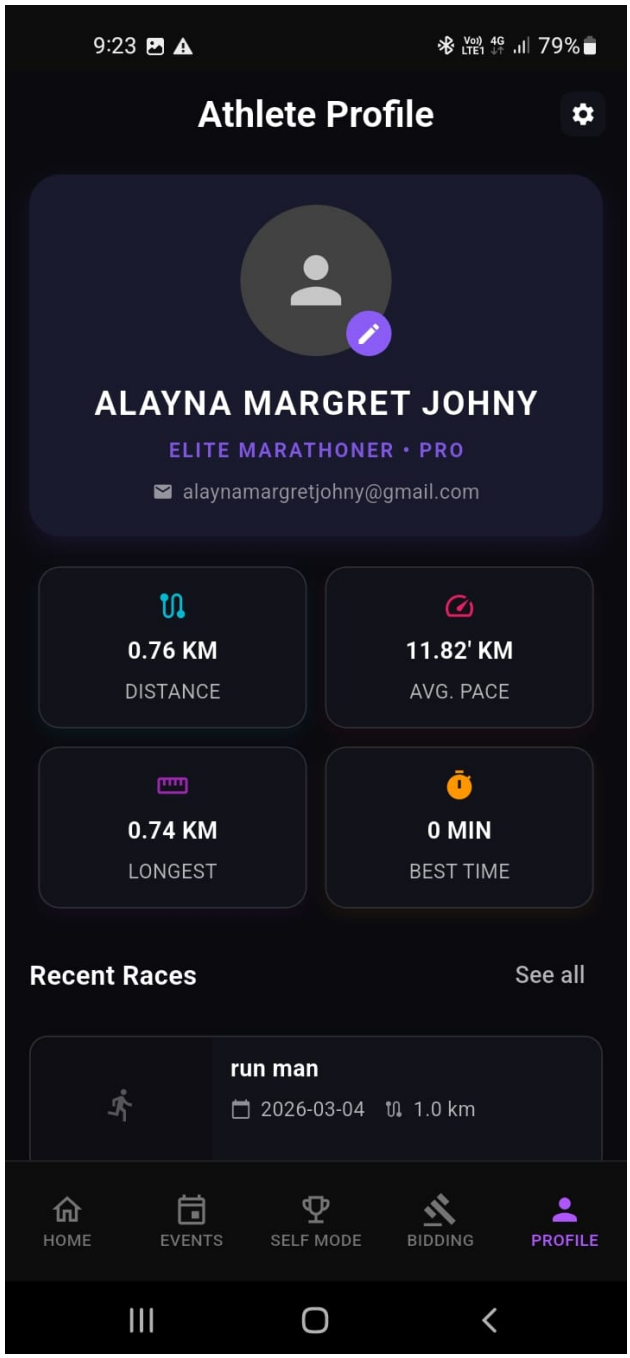


Fig. 10.8: profile page

It's an athlete profile screen showing Alayna Margret Johny's marathon stats, recent race details, and account information.

Chapter 11

SAMPLE CODES

11.1 Home

```
import 'package:firebase_auth/firebase_auth.dart';
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:thilothaman/events.dart';
import 'package:thilothaman/marathon_details.dart';
import 'package:thilothaman/bidding.dart';
import 'package:thilothaman/notifications.dart';
import 'package:thilothaman/self.dart';
import 'package:thilothaman/profile.dart';
import 'package:thilothaman/store_page.dart';
import 'package:firebase_ai/firebase_ai.dart';
import 'dart:async';



class MarathonHomePage extends StatefulWidget {
  final GenerativeModel model;

  const MarathonHomePage({super.key, required this.model});

  @override
  State<MarathonHomePage> createState() => _MarathonHomePageState();
}

class _MarathonHomePageState extends State<MarathonHomePage> with SingleTickerProviderStateMixin {
```

11.2 login

```
// lib/login_page.dart
import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:thilothaman/forgetpassword.dart';
import 'homepage.dart';
import 'signup.dart';
import 'package:firebase_ai/firebase_ai.dart';
class LoginPage extends StatefulWidget {
  const LoginPage({super.key});

  @override
  State<LoginPage> createState() => _LoginPageState();
}

class _LoginPageState extends State<LoginPage> {
  final _formKey = GlobalKey<FormState>();
  final emailController = TextEditingController();
  final passwordController = TextEditingController();

  bool _isLoading = false;
  bool _isPasswordObscured = true;

  @override
  void dispose() {
    emailController.dispose();
    passwordController.dispose();
    super.dispose();
  }
Future<void> loginUser() async {
  if (!_formKey.currentState!.validate()) return;

  setState(() => _isLoading = true);

  try {
    final cred = await FirebaseAuth.instance.signInWithEmailAndPassword(
      email: emailController.text.trim(),
```

11.3 Signup

```
import 'dart:math';
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:thilothaman/otppage.dart';
import 'package:thilothaman/services/email_service.dart';

const Color accentColor = Color(0xFF9C27B0); // Purple accent

class SignUpPage extends StatefulWidget {
  const SignUpPage({super.key});

  @override
  State<SignUpPage> createState() => _SignUpPageState();
}

class _SignUpPageState extends State<SignUpPage> {
  final _formKey = GlobalKey<FormState>();
  final nameController = TextEditingController();
  final emailController = TextEditingController();
  bool _isLoading = false;

  Future<void> _sendOtp() async {
  if (!_formKey.currentState!.validate()) return;
  setState(() => _isLoading = true);

  try {
    final email = emailController.text.trim();
    final otp = (100000 + Random().nextInt(900000)).toString();

    // Call your server to store OTP and send email
    final success = await EmailService.sendOtpEmail(email, otp);

    if (success && mounted) {
      Navigator.push(
        context,
```

11.4 Events page

```
// events_page_dark.dart
import 'dart:async';
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:google_fonts/google_fonts.dart';
import 'package:thilothaman/createmarathon.dart';
import 'package:thilothaman/marathon_details.dart';

class EventsPage extends StatefulWidget {
  const EventsPage({super.key});

  @override
  State<EventsPage> createState() => _EventsPageState();
}

class _EventsPageState extends State<EventsPage> with SingleTickerProviderStateMixin {
  String _filter = 'date';
  String _searchQuery = '';
  final TextEditingController _searchController = TextEditingController();
  late final AnimationController _listAnimController;

  // Theme colors
  static const Color violet = Color(0xFFB056FF);
  static const Color violetDark = Color(0xFF6B2C91);
  static const Color bgBlack = Color(0xFF0D0D0D);
  static const Color cardDark = Color(0xFF121212);
  static const Color cardBorder = Color(0x33B056FF); // subtle violet border

  Query<Map<String, dynamic>> _getQuery() {
    final base = FirebaseFirestore.instance.collection('marathon');
    switch (_filter) {
      case 'participants':
        return base.orderBy('participants_required', descending: true);
      case 'volunteers':
```

11.5 webstore

```
// lib/store_page.dart
// Store page with cart badge showing total quantity in carts/{uid}/items
// Dark theme, 2x2 grid, search, admin FAB, and cart badge in AppBar.
// Requirements in pubspec.yaml:
//   cloud_firestore: any
//   firebase_auth: any
//   intl: any

import 'dart:math';
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:intl/intl.dart';
import 'package:thilothaman/manage_products.dart';
import 'cart.dart'; // ensure path matches your project
import 'buyproductspage.dart'; // ensure path matches your project
// replace with your actual manage page import

class StorePage extends StatefulWidget {
  const StorePage({super.key});
  @override
  State<StorePage> createState() => _StorePageState();
}

class _StorePageState extends State<StorePage> {
  final FirebaseFirestore _firestore = FirebaseFirestore.instance;
  final TextEditingController _searchCtrl = TextEditingController();

  static const Color _bg = Color(0xFF0A0A0F);
  static const Color _card = Color(0xFF1E1E2C);
  static const Color _violet = Color(0xFF8B5CF6);
  static const Color _muted = Color(0xFF9CA3AF);

  final NumberFormat _rupee = NumberFormat.currency(locale: 'en_IN', symbol: '');
```

11.6 Self page

```
import 'dart:async';
import 'dart:convert';
import 'dart:ui';
import 'package:flutter/foundation.dart';
import 'package:flutter/material.dart';
import 'package:geolocator/geolocator.dart';
import 'package:fl_chart/fl_chart.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter_tts/flutter_tts.dart';
import 'package:http/http.dart' as http;
class SelfTrackingPage extends StatefulWidget {
  final String userId;

  const SelfTrackingPage({
    Key? key,
    required this.userId,
  }) : super(key: key);

  @override
  State<SelfTrackingPage> createState() => _SelfTrackingPageState();
}

class _SelfTrackingPageState extends State<SelfTrackingPage> with SingleTickerProviderStateMixin {
  // Run tracking variables
  final Stopwatch _stopwatch = Stopwatch();
  StreamSubscription<Position>? _positionStream;
  Position? _lastPosition;
  double _totalDistance = 0.0;
  bool _isRunning = false;

  // TTS variables
  final FlutterTts _tts = FlutterTts();
  bool _isSpeaking = false;

  // Coach variables
  final TextEditingController _coachController = TextEditingController();
  String _coachReply = '';
```

11.7 Bidding

```
// bidding_page_search.dart
import 'dart:async';
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:thilothaman/biddingdetail.dart';

class BiddingPage extends StatefulWidget {
  const BiddingPage({super.key});

  @override
  State<BiddingPage> createState() => _BiddingPageState();
}

class _BiddingPageState extends State<BiddingPage> with SingleTickerProviderStateMixin {
  final FirebaseFirestore _firestore = FirebaseFirestore.instance;
  final User? _user = FirebaseAuth.instance.currentUser;
  late final AnimationController _pulseController;

  bool _showFinished = false; // false => Upcoming, true => Finished

  // Search
  final TextEditingController _searchController = TextEditingController();
  Timer? _searchDebounce;
  bool _isSearching = false;

  // Search results: list of marathon docs and their matching product docs
  List<_SearchResult> _searchResults = [];

  @override
  void initState() {
    super.initState();
    _pulseController = AnimationController(vsync: this, duration: const Duration(milliseconds: 900))
      ..repeat(reverse: true);

    _searchController.addListener(_onSearchChanged);
  }
```

11.8 profile page

```
// self_profile_page_full.dart
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'package:image_picker/image_picker.dart';
import 'package:firebase_storage/firebase_storage.dart';
import 'package:thilothaman/about.dart';
import 'login.dart';

class SelfProPage extends StatefulWidget {
  const SelfProPage({super.key});

  @override
  State<SelfProPage> createState() => _SelfProPageState();
}

class _SelfProPageState extends State<SelfProPage> {
  final uid = FirebaseAuth.instance.currentUser!.uid;
  String? photoUrl;

  // Theme colors
  static const Color bg = Color(0xFF0A0A0F);
  static const Color card = Color(0xFF12121A);
  static const Color accent = Color(0xFF8B5CF6);
  static const Color accentSoft = Color(0xFFB56CFF);

  @override
  void initState() {
    super.initState();
    // Load initial photoUrl
    FirebaseFirestore.instance.collection('users').doc(uid).get().then((snap) {
      final data = snap.data();
      if (data != null && mounted) {
        setState(() {
          photoUrl = data['photoUrl'] as String?;
```

Chapter 12

CONCLUSION & FUTURE SCOPE

The Stridex Marathon Management System is designed to simplify and streamline the organization of marathon events by integrating essential features such as participant registration, sponsorship management, event coordination, and real-time tracking. It enables organizers to efficiently handle multiple aspects of event management in a structured manner, reducing manual workload and improving overall coordination[1]. At the same time, participants benefit from an easy-to-use platform where they can register, monitor their runs, and analyze their performance, resulting in a more engaging and user-friendly experience.

Looking ahead, the system can be further enhanced by incorporating advanced features and improving existing modules. Future developments may include upgrading the Self Run module for more accurate tracking, integrating wearable fitness devices, enabling secure online payment options for registrations and sponsorships, introducing live leaderboards for participants, and enhancing real-time GPS tracking capabilities[1],[17]. These improvements will make the system more efficient, reliable, and adaptable to modern technological needs, ultimately providing a better experience for both organizers and runners

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